

Variance of a Population:

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

each data point

N

mean of the population

Number of total data points in the population

Variance of a Sample:

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

mean of the sample

one less than the number of data points in the sample. This is known as Bessel's correction to help correct the bias in the sample.

Standard Deviation of a Population:

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}}$$

Standard Deviation of a Sample:

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$